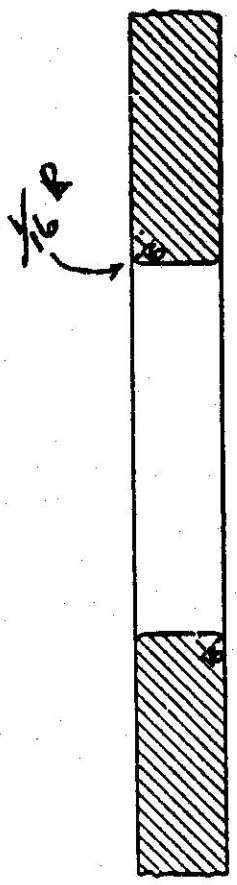
center. From this portion two longitudinal strips approximately 1″ wide are to be cut for bend test. One of these strips is to be bent against the outside surface and the other against the inside surface of the flue. In conducting the bend test the specimen should be placed in a vise with the edge of the weld 1/16″ to 1/8″ above the edge of the vise, the welded portion protruding from the vise. The extended end of the specimen is then to be bent the required direction over the jaws of the vise through an arc of 60 to 70 degrees.

(f) Under these tests a weld will be considered 100 percent if the specimen can be bent without showing any parting of the edge of the scarf. Welds which part slightly at the edge of the scarf with the scarf showing bright metal will be considered satisfactory. Welds which readily open or show oxidized non-fused areas are considered unfit for service. The reasons for defective welds should be investigated at once and corrected.

3. **Ferrules:** Ferrules for boiler flues, including those for superheater flues, must be made of soft copper, which is .075 inch thick. The length and the outside diameter of the ferrules must conform to dimensions shown on standard tracings. Where not shown on tracing, the outside diameter should be made the same as the outside diameter of the flue and the length should be made 1/32 inch greater than the thickness of the flue sheet.

4. **Preparation of Fire Box Sheets:** (a) The diameter of holes in fire box sheets for flues and arch pipes must conform to dimensions shown on standard tracings. Where tracings do not show dimensions, the diameter of the holes must be the same as the outside diameter of the flues. All holes in fire box sheets for flues and arch pipes must have the inside and outside edges rounded to 1/16″ radius as indicated by Sketch "A".



SKETCH "A"

(b) New firebox flue sheets must be thoroughly cleaned on fire side by either sand blasting or grinding to remove all scale before being applied and old sheets must be cleaned in like manner after the flues are removed in order to facilitate the electric welding of flues.

5. **Reaming of Flue Sheets:** When a flue hole in a fire box sheet becomes 1/8 inch out of round, it must be reamed. The reamed hole must not be larger than 3/16″ over standard diameter of flue hole.

6. **Preparation of Smoke Box Flue Sheets:** Unless otherwise shown on standard tracings, the diameter of the holes in the smoke box flue sheet must be 1/16 inch larger than the flues whose diameter is 3 1/4 inches or less, and 3/32 inch larger than the flues whose diameter is 3 1/2 inches or more. All holes in smoke box flue sheet must have the inside and outside edges rounded to 1/16 inch radius, as indicated by Sketch "A".

7. **Preparation of Fire Box End of Flues:** Fire box end of flues must be swaged down with a slight taper toward the end of the flues so that the diameter at the end will be .150 inch less than the diameter of the flue hole in order to permit the end of the flues to fit the ferrule neatly after the ferrule has been tightened in the flue hole. The swaged ends of all flues, except the large 3 1/2 and 5 1/2 inch superheater flues, must be at least 1 1/8 inches long as shown in Figure 3. The swaged ends of large superheater flues must be at least 3 inches but not more than tracing size in length, as shown in Figure 10.

8. **Preparation of Smoke Box End of Flues:** Flues should be cut to length and the smoke box end opened hot to such a size that when placed in boilers they will be a snug fit in front flue sheet holes. This should not be done to flues which are to be transferred in the boiler. The 5 1/2 inch superheater flues need not be opened, except to take care of enlarged holes. For method of swaging ends of flues in the front